Design Alternatives and Create Prototypes

Emin Zerman

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Project Work – So Far

- Form groups of 3-5 students.
- Come up with a project idea. What problem or need are you going to solve?
- Think about who your target group is!

- Performed data collection (interview/observation/survey)
  - Data in the form of, for example, notes from interviews/observations or survey responses.

- Develop use cases, scenarios and/or personas .
- List functions that should be in your design.
- Prioritize and focus on the most important functions

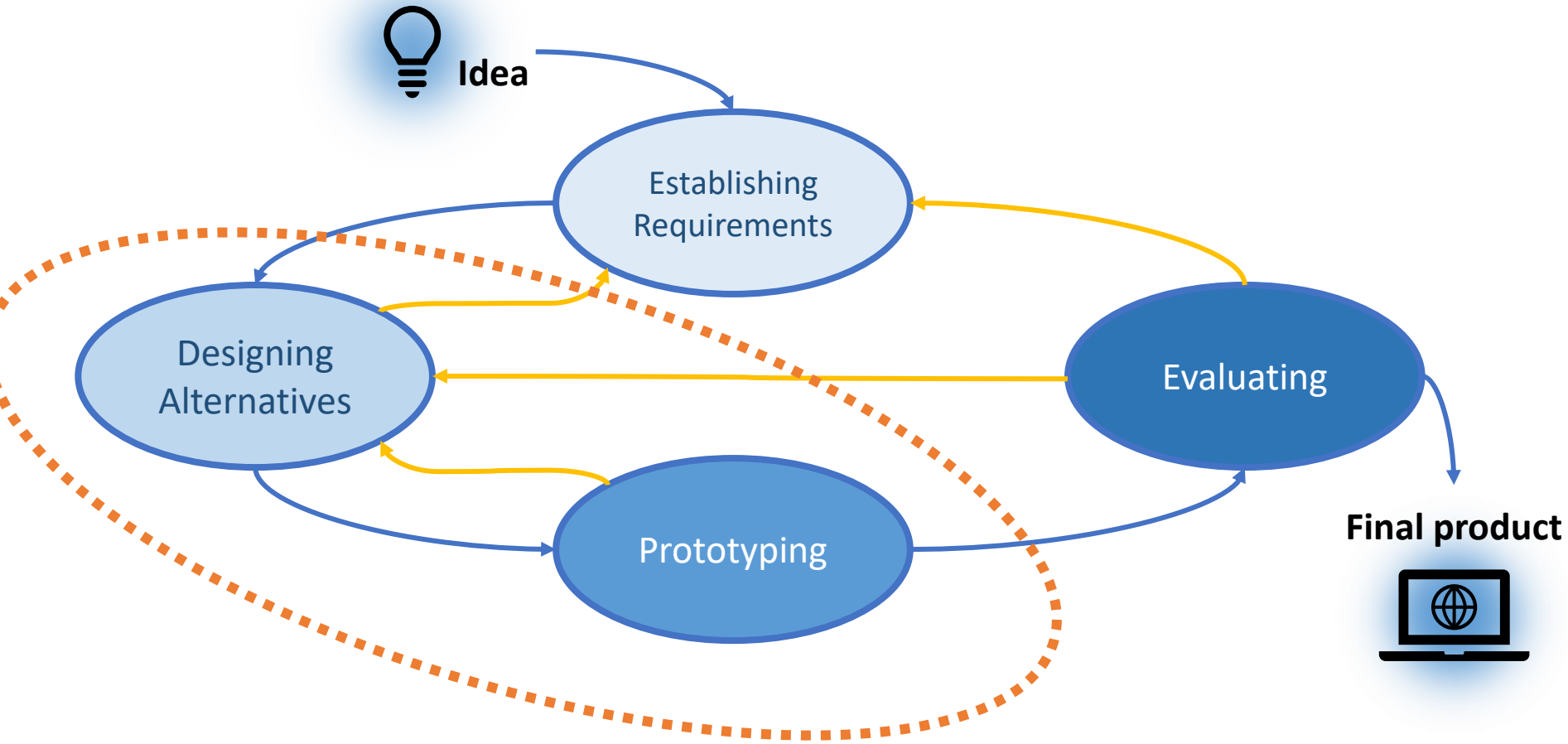
Timeline

	Mon	Tue	Wed	Thu	Fri
v38	Project start (C015 and C016, ALC) →				Data collection (M209)
v39	→		Data analysis (M209)	→	
v40	→				
v41	Prototypes (C015 and C016, ALC)			Usability tests (M209)	
v42					
v43					
v44			Presentations	Presentations	

What are the biggest challenges so far?

Let’s do an anonymous data collection with Menti!

Design Process



Preece, J., Sharp, H., & Rogers, Y. (2015). **Interaction design: beyond human-computer interaction**. John Wiley & Sons.

Can you answer the following questions?

Aspect of the concept	Design questions
Idea	What is the product or service for? *-- (an identifiable whole from the perspective of users or other stakeholders)* What is the name of the concept?
Who	Who is the product or service for? Who is affected by it (short term/long term)?
What	What is the main function? What is the most important content?
When and where	In which social contexts and in which physical environments can the product or service be used?
How	What principles does the product or service follow? What qualities are sought? What types (or kinds) of design options are available?
Why	Why is the product or service used? What problem or question does it answer?
Consequences	How are people, the environment, society, or the environment affected by it?

From the book *Interaction design and UX - about creating a good user experience* (Mattias Arvola , 2021)

Concept Idea

When you have your overall intentions clear, it's time to generate ideas and design concepts. Every concept idea should have

a name – for example " Menomap ",

a main purpose – for example "Offer interactive maps to hikers via various channels,

design principles – for example "Maximize the map experience in simplicity, utility, challenges and fun."

(Arvola , 2021)

Examples of techniques for concept generation

- Brainstorming
  - **No criticism** – ALL ideas are welcome
  - Get many ideas, **focus on generating ideas**
  - Combine and improve on the ideas
  - Better to have many short sessions than a long session

- Brainwriting (Method 635)
  - A more systematic way of brainstorming
  - More detailed in the next slide

- Role play

- Monitoring the world for inspiration

Brainwriting - Method 635

- Gather 6 participants, *(hence the number 6)*
  - who in the first round begin by getting familiar with the challenge
- Participants individually write down 3 ideas *(hence the number 3)*
  - In a fixed time (~5 minutes) – might be extended if necessary
- The three ideas are passed on to the next person at the table, who reads through them and adds three new ideas or modifications to the previous ideas.
- Do this a total of 5 rounds. *(hence the number 5)*

Brainwriting - Method 635

Challenge / Question / Problem is:			
Round	Idea #1	Idea #2	Idea #3
1			
2			
3			
4			
5			
6			

Brainwriting - Method 635

Advantages:

- Refining good ideas
- Avoiding the problem of some participants dominating, (while others disappear)
- It makes it easier to see who came up with a certain solution.

Disadvantages:

- Compared to brainstorming, is that some of the dynamism is lost.

Exercise: Try to answer these questions

Try for 15 minutes & hand-in your answers
If you cannot answer a question, Come up with an explanation to what is lacking

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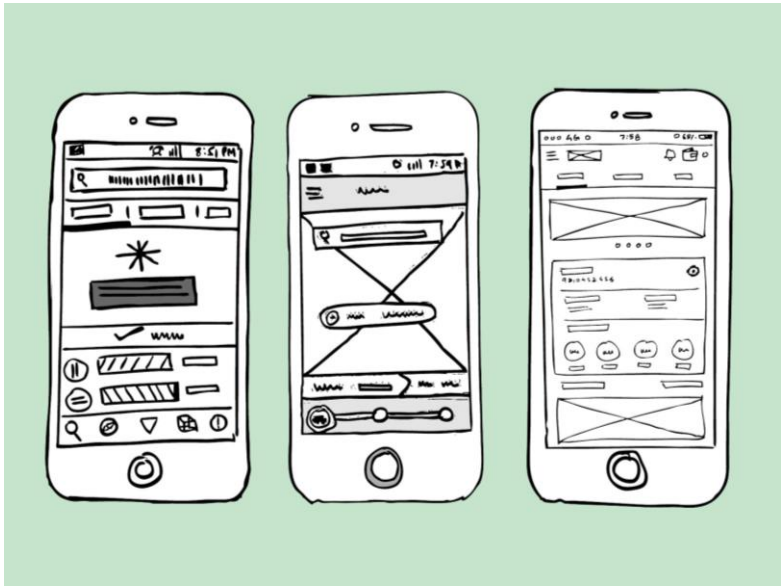
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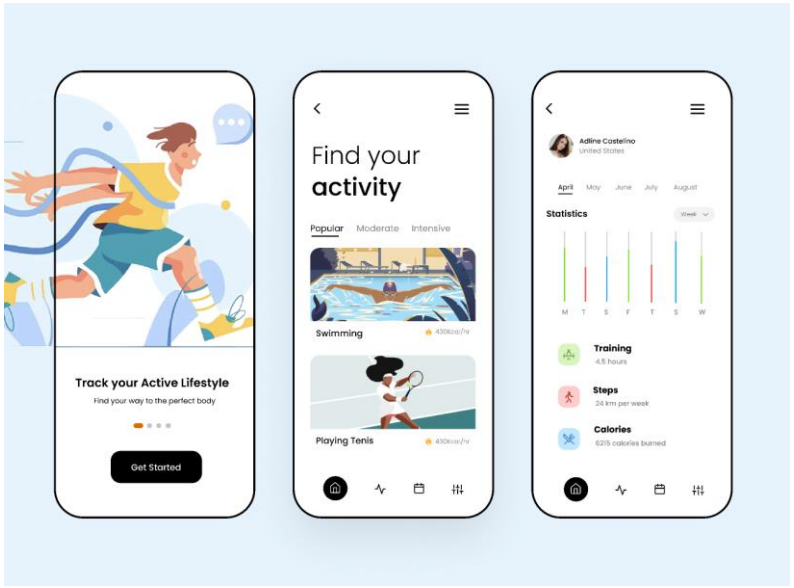
Interface

Which type of interface best solves your problem?

- Graphical (GUI),
- Voice,
- Touch,
- Multimodal,
- Virtual Reality,
- Augmented Reality

Fidelity in Prototyping





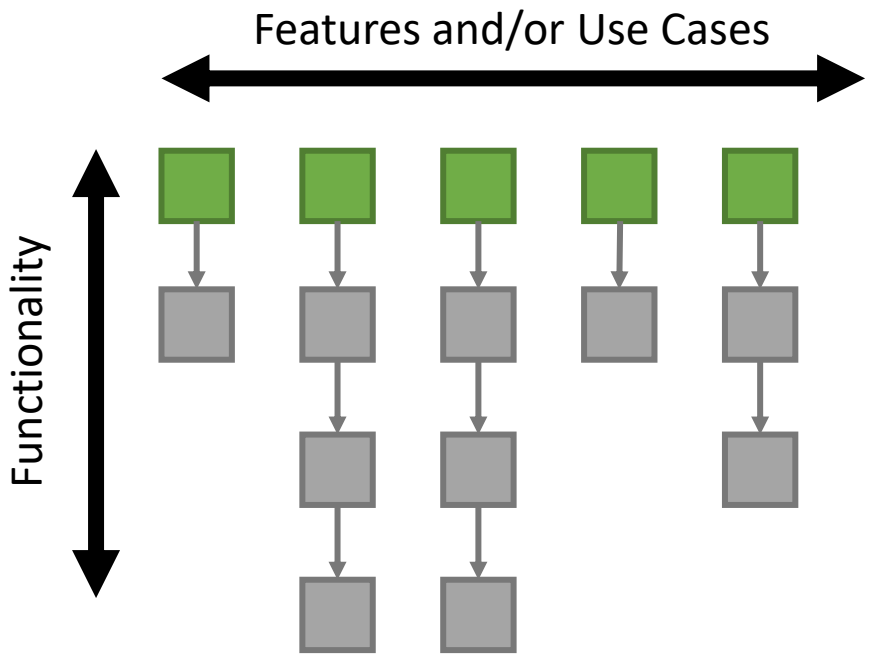
Low-fidelity (Lo-fi)

- Can quickly design different options
- Good for early testing

High-fidelity (Hi-fi)

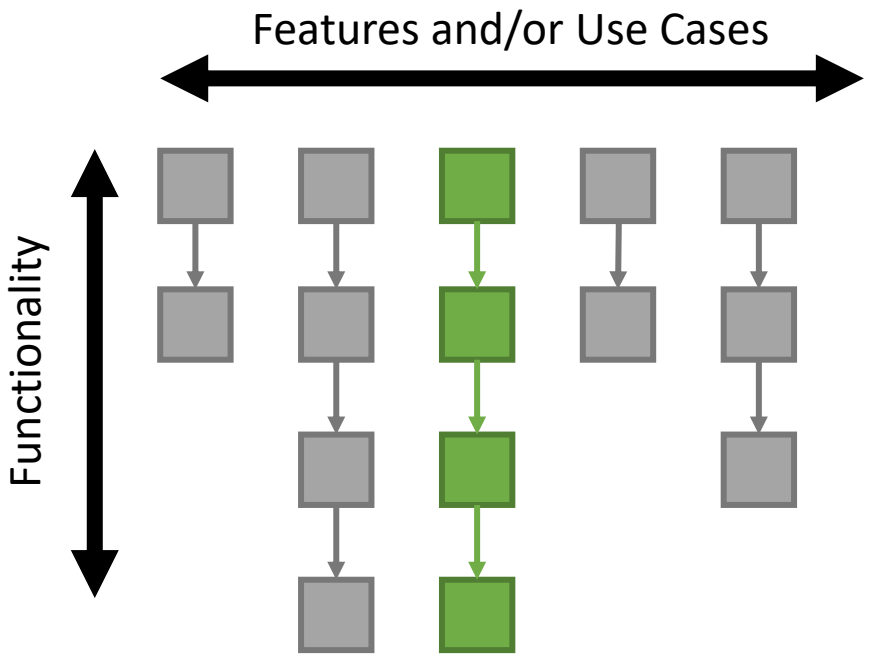
- Interactive
- Easier to test

Prototype design



Horizontal design

- Many functions with few details

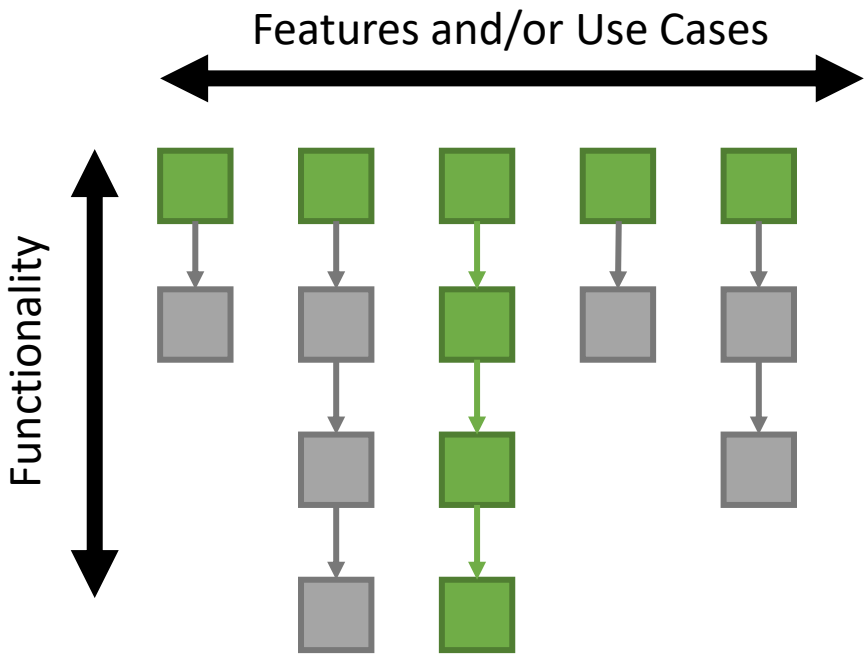


Vertical design

- Few features with a lot of detail

Redrawing is based on: Fox, D., Sillito, J., & Maurer, F. (2008, August). Agile methods and user-centered design: How these two methodologies are being successfully integrated in industry. In Agile 2008 Conference (pp. 63-72). IEEE.

Prototype design



T design

- Many functions with few details + Few features with a lot of detail

Prototype design

Conceptual design

- A conceptual model is
  - "a **high-level description** of how a system is organized and operates".

  It can thus be seen as an overview map of how something should work.

- Interaction types
  - (instructional, conversational, manipulative, exploratory)

Concrete design

- A more complete design
  - Colors, graphics, icons, buttons, layout

- Apply design principles

Exercise !

Have enough to go with prototyping?



Continue to work with data collection and data analysis



- **List** functions that should be in your design and prioritize
  - Feel free to try the function analysis method
  - Try brainstorming and brainwriting
- **Find** what suits your prototype
  - Horizontal or vertical design?
- **Decide** what kind of interaction you should have

- **Create a lo-fi prototype**
- **Create a hi-fi prototype**

We'll go through usability testing in the next exercise.